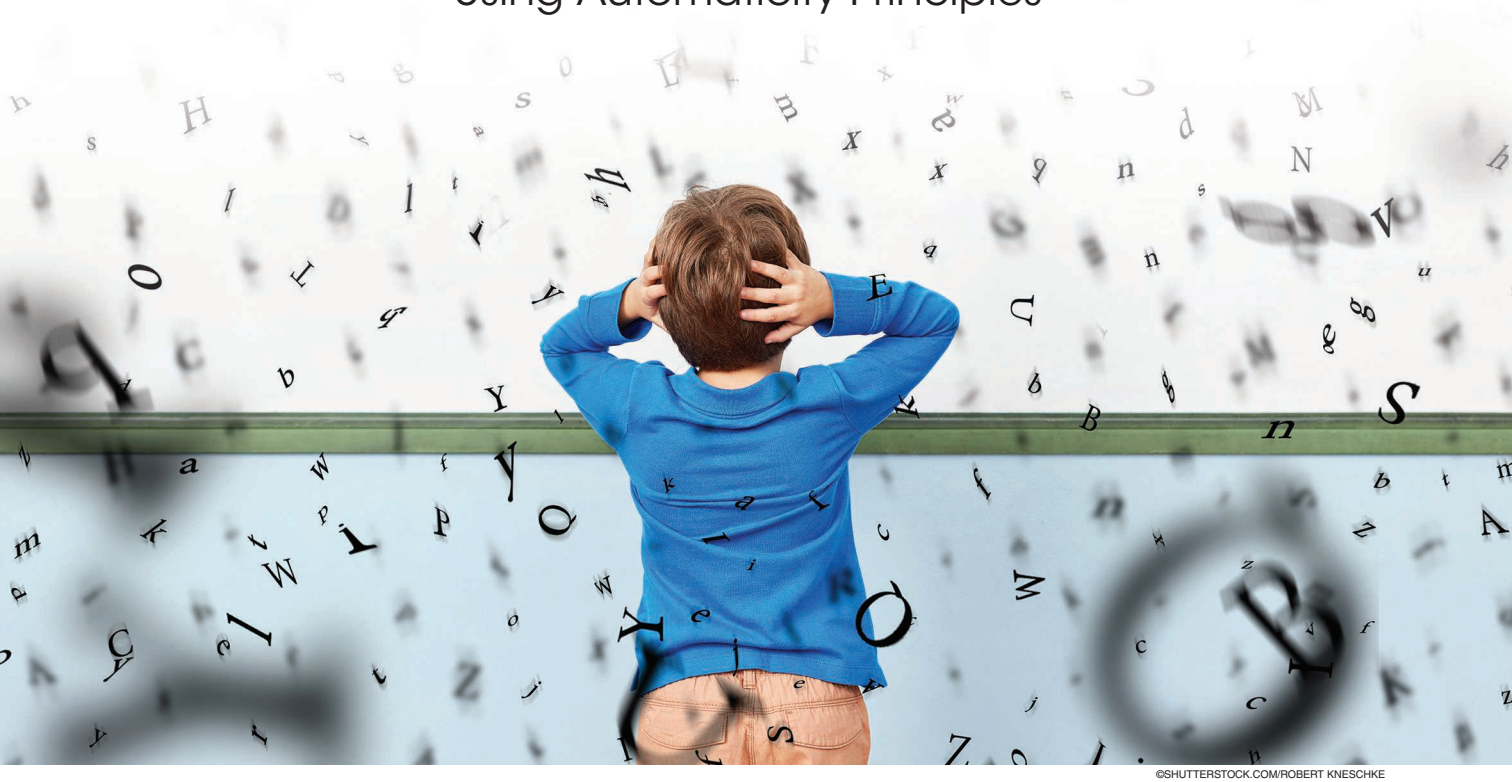
Intelligent Games for Learning and the Remediation of Dyslexia

Using Automaticity Principles



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by Hossein Jamshidifarsani, Samir Garbaya, Theodore Lim, and Pierre Blazevic

The ability to read has become an indispensable skill in modern ages, and any reading difficulty, such as dyslexia, can seriously impair the aspirations of the individual. Orthographically opaque languages such as English lay a heavy burden on learners. In this article, a gamified intervention program for the remediation of dyslexia is proposed for opaque orthographies. Current technology-based approaches of reading acquisition in the literature lack sophistication in terms of training design, game design, and adaptivity. This approach is based on the principles of automaticity acquisition and the gamification of learning as well as intelligent instruction. For the latter, an optimization model is proposed to maximize the educational value of each training session while respecting the capabilities of each individual.

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The analysis of the data collected from the two usability tests of the system reported in this article provided valuable information for the improvement of the gamified content, the interaction design, the length of playing sessions, and the situational motivation scale. The optimization model integrated into the games selected the content with the highest value (marked by the frequency level of words) and the lowest difficulty (marked by the word length) most often, and the neighboring content was selected less. However, the data from error rates committed by the users showed that for an accurate prediction of word difficulty, a combination of lexical information is necessary as neither word length nor word frequency could achieve that alone. Overall findings, along with the feedback from the users, validate the game design as well as the optimization model integrated inside the games.

Technology-Based Remediation of Dyslexia

Dyslexia is the most prevalent form of learning difficulties, with researchers differing on its prevalence, ranging from 5 to 20% [1]. It can have serious negative impacts on a person's self-esteem and academic aspirations [2]. Some studies have pointed out that for treating dyslexia, the instruction duration should be between 80 and 100 h, while healthy peers need 30 to 60 h [3], [4]. This extra amount of needed instruction is often not provided in schools.

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Ubiquitous technologies can potentially mitigate this lack of sufficient instruction at school. Technology-based approaches offer several advantages to traditional instruction. They can personalize the instruction for each individual as well as be accessed at any time and in any location. Therefore, individual training time can be increased and the instructor's time can be freed up for engaging in other tasks [5]. Furthermore, by applying the gamification concept, instruction can become more engaging and increase the adherence of the students to the intervention program [6]. However, despite the aforementioned advantages, constructing such approaches requires a multidisciplinary effort and deep knowledge of the targeted instructional topic and game design theories as well as learning and training theories [6].

Technology-based in-home remedial programs can offer all of the advantages mentioned previously, but only if they are well designed. A crucial aspect of these programs is their ability to adapt the learning material to the capabilities of the users. If the materials are too difficult, the user will get frustrated or anxious, and if the learning materials are too easy, the user will feel bored [7]. In addition to the adaptive difficulty level, the program should also be able to provide the content with maximum educational value. These intelligent characteristics are essential for any in-home interventional programs.

In this article, a reading intervention program was designed to target young dyslexic children who have to learn orthographically opaque languages (in this case, English). To promote automaticity in decoding, the training principles of automatic skill acquisition were taken into account. An intelligent component was created to tailor the educational content of the tasks to the capabilities of each learner as well as maximizing the educational value of each training session. In addition, other adaptive features, such as adaptive speed and adaptive rewards, were integrated into the system. Furthermore, the gamification principles as well as motivational theories of self-efficacy and self-determination were used to design a playful and game-based environment for the reading tasks.

Related Works and Problem Statement

There are some studies in the literature that tried to gamify interventions in reading instruction. *GraphoGame* (also known as *Ekapeli* or *Literate*) [8] is a gamified phonics system for teaching the relationships between speech sounds and their written counterparts. A Balanced Reading Approach for Children Always Designed to Achieve Best Results for All is a web-based gamified tool focusing on teaching phonics, and its effectiveness has been subject to a meta-analysis [9]. van de Ven et al. [10] have tested a multicomponent mobile game called *Lett. Prince* (*Letterprins*). Messer and Nash [11] evaluated a gamified computer system called Trainertext that is based on visual mnemonics to teach and train grapheme–phoneme relations. A multicomponent reading game called *Meister Cody-Namagi* has been recently evaluated [12]. It covers phonological awareness and phoneme–grapheme mapping as well as word reading skills. Another study evaluated the effectiveness of an augmented reality game focusing on reading comprehension [13].

A recent review [14] analyzed technology-based reading interventions and identified several issues with current practices. A few of these issues related to this study include the following.

1) Working on the automaticity aspect of reading is generally neglected in elementary schools, and it is even more neglected in earlier grades and for languages with opaque orthography.
2) Fluency approaches were, on average, shorter, which does not correspond to the extensive practice time required for acquiring it. In addition, fluency interventions were not gamified. Gamification can allow users to practice for longer durations, which is vital for fostering fluency.

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3) Vocabulary interventions were overlooked; this is more problematic for opaque orthographies in which decoding depends a lot on making analogies based on the current vocabulary knowledge.

Furthermore, the currently reported models in the literature that target creating content for reading instructions are rudimentary and not adaptive. For example, one way is to randomly present the words to the user and then omit the correct answers from the pool of the possible words, and the other method is to keep correct answers in the pool and omit the mistakes [15], [16]. Nevertheless, these approaches are basic and do not consider the difficulty level of the words, and, in addition, the values of all the words in the pool are always considered the same or as binary. Finally, the COVID-19 pandemic, which has led to unprecedented global lockdowns, has highlighted the value of intelligent home-based educational tools [17].

The Proposed Approach

The sections that follow describe the implemented approach, starting with the system architecture.

The Architecture of the System

For mitigating the issues raised in the "Related Works and Problem Statement" section, and to propose a solution for the reading acquisition of languages with opaque orthographies, a gamified intelligent approach for the remediation of dyslexia is proposed. It focuses on fostering the automaticity of decoding and recognition at the lexical level (explicitly) and sublexical level (implicitly) as well as providing a basis for extensive vocabulary instruction. A home-based approach is proposed to provide sufficient training time required for the development of automaticity in decoding and the extensive vocabulary instruction needed for influencing generalized vocabulary knowledge.

Gamification is proposed to increase motivation, engagement, and adherence of the users to the training program, which is supposed to be used extensively. Finally, an intelligent learning system based on an optimization model is developed to make the home-based intervention possible without the need for the physical presence of the instructor. This model will maximize the value of each training session while respecting the ability level of the user. Each of these proposed aspects of the training program is explained in more detail.

Reading Tasks Designed for the Intervention

In this study, four different reading tasks were created to address the problems raised in the previous sections. These tasks were divided into the two modules of fluency and vocabulary. The fluency module contains three tasks. The first two tasks are called *accelerated word decoding* (*AWD*) and *accelerated word–sound recognition* (*AWSR*). These two reading tasks act as the opposite of each other. In the first task (AWD), the word is presented whereupon the user should decode it and read it aloud as fast as possible. Speech recognition technology is used to decide whether the pronunciation was accurate or not. Speech recognition engines provide a parameter called the "confidence level." Different levels allow for a looser or stricter recognition. As choosing the right level is a matter of trial and error, this parameter was put on the cloud to be able to change it immediately in accordance with the feedback from users.

However, in the second task (AWSR), the user hears the pronunciation of a word through text to speech (TTS) technology, and then he/she should recognize the pronounced word among four possible choices. Depending on the level of the users, the three distractors can be either random words for novices or orthographically similar words for more advanced users. Despite the higher clarity of human recorded speech, TTS is an agile and practical tool for dealing with thousands of words.

The term *accelerated* is used to signify the time pressure aspect that pushes the user to decode or recognize the words increasingly faster. Time pressure is one of the important attributes of automaticity training [18], [19]. However, one key factor in automaticity is the consistency of the components, rules, or stimuli responses that have to be automatized [20], [21]. In languages with opaque orthography, the consistency between letters and sounds does not always exist; hence, the reader should find consistency at larger levels, and, in some cases, consistency can be found only at the whole word level or, even worse, in the cases of heteronyms, context is required to cue that consistency.

Another key factor in automaticity acquisition is the contiguity of stimulus response or rules or regularities [22]. This matters a lot in implicit learning, where the learner should detect the rules or regularities [23]. It means that consistent rules and regularities should be proximal to each other in terms of time and space to be detected by the individual. Finally, another key attribute in automaticity acquisition is to start with blocked and massed practice.

In the two games that were designed to target automaticity, all of the aforementioned factors were taken into account. The words were chosen from 334 word lists of phonograms (extracted from the book *Phonics from A to Z: A Practical Guide* [24]). The words with the same phonograms were grouped together to be practiced in blocks. This also increases the contiguity of sublexical regularities, which in turn increases the probability of implicit acquisition. In addition, the time pressure technique was used to push the user to decode increasingly faster, which can accelerate the process of automaticity acquisition. Practicing these two tasks in the massed format are expected to foster lexical automaticity explicitly and sublexical automaticity implicitly.

The third fluency task is called *accelerated phrase reading* (*APR*), which focuses on increasing the reading rate. This task is based on a text-fading method developed by

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Breznitz and Nevat [25]. It presents a phrase to the user, and the text starts to disappear letter by letter from the beginning. After the text has totally disappeared or the user has signaled that he/she has finished reading, a four-choice comprehension question is presented to the user. Depending on the answer, the text-fading speed will be increased or decreased. This keeps the users on the edge of their reading rate and pushes them to read the phrases faster.

Finally, the reading task in the vocabulary module is a cloze question called *accelerated word meaning recognition* (*AWMR*). It represents a phrase with one word missing, and the user has to select the appropriate word and ignore the distractors. This task is also designed to both improve the vocabulary of the user as well as increase the automaticity in word decoding and sentence reading.

The Gamification of the Reading Tasks

Learning to read can be a hard and laborious task, especially when a child is underachieving and feeling the pressure of lagging behind. Having difficulty in learning to read can lead to demotivation and discouragement for engaging in reading activities, and this can further widen the gap between achieving and underachieving readers [26]–[28]. For poor readers, reading activities by themselves do not induce enough motivation to encourage persistence and adherence in children, and instructors need to target both reading skills and motivation [29]. Hence, that is why gamification is proposed as a mechanism to increase motivation, engagement, and persistence in the process of reading acquisition.

Gamification has been defined as "using game design elements in nongaming context" [30]. It has been shown to make school lessons more interesting and motivating for the students [31]. It can positively affect the learner's attitude and behavior toward learning tasks [32]. In addition, it can be used to improve the learner's engagement [33], [34] as well as their autonomy and creativity [35]. In addition, it is found that, despite increasing cognitive load, it can lead to higher achievement among students [36].

Through the use of gamification principles, the four reading tasks designed for this system were transformed into four minigames. A thematic was chosen for all the minigames to make them look coherent. As the system targets young children, this thematic was chosen to be the four holidays of Christmas (AWD), Halloween (AWMR), Easter (APR), and Thanksgiving (AWSR). Various game elements were incorporated into this gamified system. To summarize, they cover reinforcements (continuous, intermittent, and variable as well as positive and negative), goals and progress bars (proximal and distal), feedback (immediate, concurrent, and corrective), and finally, scores, levels, and leaderboards (micro and macro). In addition, motivation theories such as self-efficacy (mastery experience, vicarious experience, and verbal persuasion) and self-determination (autonomy, competence, and relatedness) [37]–[39] were used as guidelines to design the gamified system. The architecture of the system is found in Figure 1.

The Optimization Model of the Intelligent Instruction

The process of skill acquisition should be adapted to one's individual needs and characteristics. If the training task is too difficult, it will produce frustration, and if the task is too easy, it could become boring to the user [7], [40]. According to the flow theory [7], the balance between skills and challenges is the key to increasing motivation and focus. In real learning processes, normally, it is the role of a teacher or a tutor to determine the appropriate educational content and its suited difficulty level. The teacher should offer the educational content that provides the most value and make sure that the tasks remain challenging without making them too difficult.

For designing an instructional tool dedicated to home use, it has to be sufficiently intelligent to eliminate the requirement of a teacher's intervention. Maximizing the educational value of a training session while keeping the difficulty level of the session below the abilities of the learner is an optimization problem. The knapsack problem is an optimization method used in previous research on education; it is appropriate for modeling the reading instructional tasks presented in this article [41]. In the knapsack problem, for a given set of items, each with a weight and value, we should choose the items that offer the best total value while not surpassing the weight limit. In the context of a reading program, every word or question has a certain weight that is defined by its level of difficulty. The more difficult words will have higher weights. Then, each word or question should have a certain value. Since the objective function of the knapsack is to maximize the total value of the selected items, an item with a higher value has a greater chance of being selected.

The constraint about the difficulty of the selected items makes sure that the set of selected items (words or questions) respect the defined weight limit. This model includes an upper weight (UW) for the upper limit and a lower weight (LW) for the lower limit. The UW limits the difficulty of the content to avoid frustration, and the LW limits its ease to avoid being boring. An additional constraint maintains the number of selected items within the defined boundaries. In this model, the upper number (UN) is the upper limit on the maximum number of items that can be selected, and the lower number (LN) is the lower limit. The objective of this constraint is to make the amount of the content of the sessions uniform; hence, the sessions are all the same length. The third constraint limits the number of presentations (P) for each word or question to a fixed maximum value so that no words or questions would be presented more than that maximum.

The starting value of each word was derived from the frequency of the word in the language (retrieved from Wordnik

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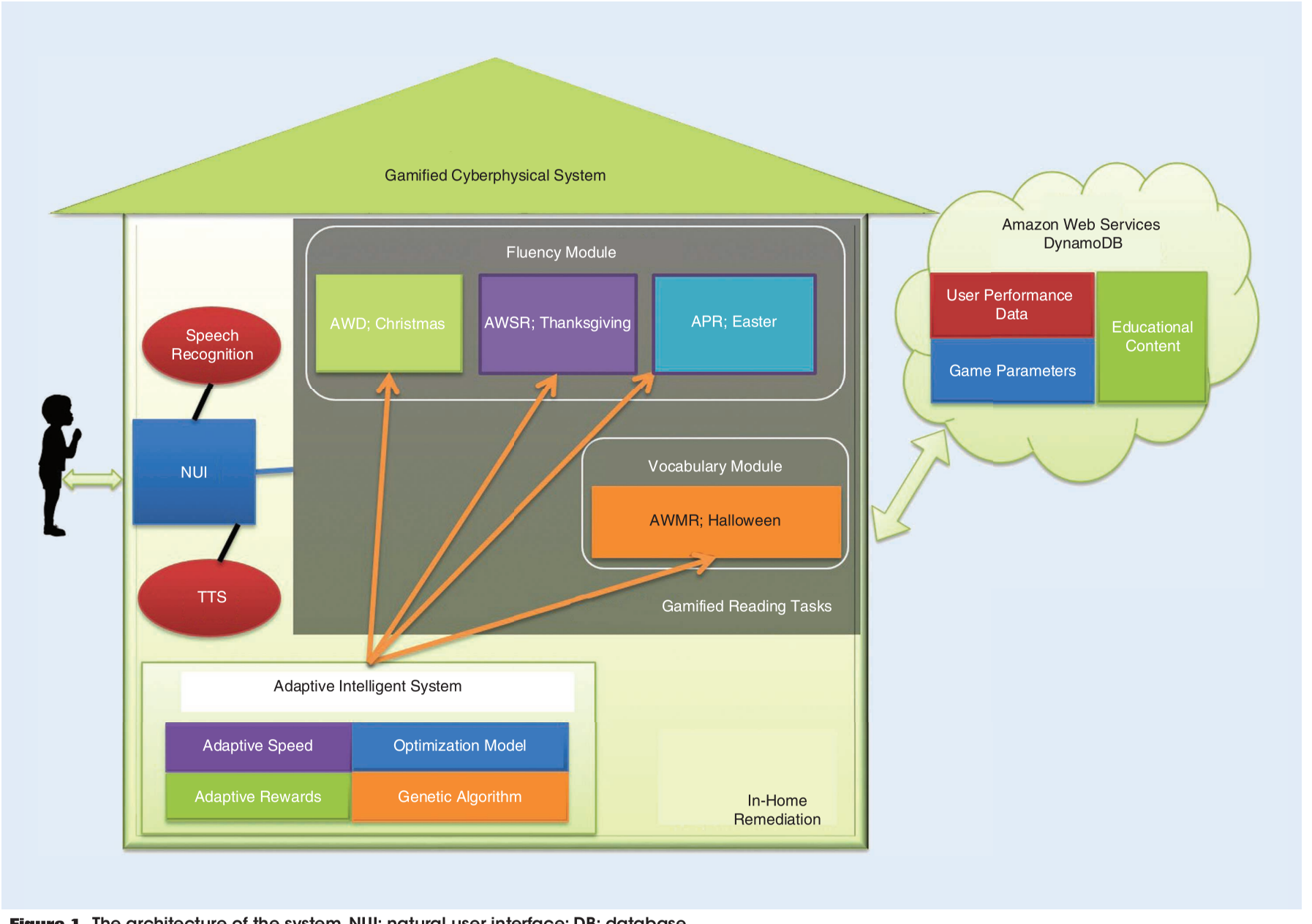


Figure 1. The architecture of the system. NUI: natural user interface; DB: database.

API [42]). Since the frequencies were based on the occurrence count of the words in bodies of texts, the numbers could be greatly different. For controlling this, the frequencies were classified into 53 categories, and numbers ranging from 1 to 53 were used for marking the value of each word. As stated previously, these values were only the starting point, and after each session, they will be updated. If a word has already been shown to the user who provided a correct answer, its value will be decreased by a certain amount so that the probability of presenting the same word again will be decreased. Similarly, if a word has been presented and the user could not provide a correct answer, its value will be increased so that the probability of presenting the same word again will be increased. The amount of these adjustments can also depend on the response time of the user.

The difficulty level of each learning material is presented by the weight parameter in the optimization model. Various attributes, such as word length, frequency, or the consistency of the phoneme–grapheme mapping, can be used for the difficulty level of a word. Here, it was decided to choose the word length (number of letters) for the weight of each word. Then, each time, depending on the performance of the user in the previous training session, the maximum and the minimum weight limits (UW and LW) will be adjusted for the next training session. If the session was too difficult for the user, the UW and LW can be lowered, and, conversely, if the session was too easy, those parameters can be increased to make the next session more difficult. Even the length of sessions can be increased and decreased depending on the level of the user and by modifying the LN and UN values.

In addition to the optimization algorithm solved by the system each time to determine the training content, various adaptive components were created into the games. Notably, the speed of the games and the rewards were adapted based on the answers of the users and their response time, respectively. The optimization model that was integrated into the games can be presented by:

Table 1. The variables and parameters of the optimization model.

x_i	Item i with a value of zero or one
v_i	The value of the item i
w_i	The weight of the item i
p_i	The number of presentations of the item i
LW	The minimum allowed weight
UW	The maximum allowed weight
LN	The lower limit on the number of selected item
UN	The upper limit on the number of selected items
P	The upper limit on the presentation count for each item

The objective of maximizing the value of each training session includes:

$$\text{Maximize} \sum_{i=1}^{n} v_i x_i, \tag{1}$$

which is subject to the constraint for the difficulty level of each training session:

$$LW < \sum_{i=1}^{n} w_i x_i \leq UW, \tag{2}$$

the constraint for the number of items in each training session:

$$LN \leq \sum_{i=1}^{n} x_i \leq UN, \tag{3}$$

and the constraint for the number of presentations for each piece of reading material:

$$\forall_i \; p_i x_i \leq P, \tag{4}$$

where

$$x_i \in \{0, 1\}. \tag{5}$$

Table 1 presents the variables and parameters of the optimization model, where x_i is the variable and the other elements are the parameters of the model.

Experimental Procedure

Two usability studies were carried out to test the functionality and usability of the games. The two studies were separated by a period of four months to allow for conducting an iterative design of the system. For the first study, 15 participants were selected from a group of English as second language learners who volunteered to take part in the functionality testing study. Sixteen other participants who were native English speakers were involved in the second study. The two samples comprise 18 males (58%), with ages ranging between 7 and 50 [mean (M) = 19.4, standard deviation (SD) = 8.9]. Four of them were left-handed (13%), and eight wore glasses (26%). Each of the volunteers played the games individually, and they were asked to play each game at least once.

A combination of subjective data (questionnaires and written feedback) and objective data (player performance and engagement) was used to evaluate various aspects regarding the behavior of the users and the functionality and interactivity of the games. A system usability scale (SUS) [43] was used to evaluate the usability of the gamified reading system, and a short version of the situational motivation scale (SIMS) [44] was administered to evaluate the perceived motivation of the users.

Data Analysis and Results

The first usability study was carried out mainly to test the functionality and interactivity of the games. The received written feedback was valuable in building the second design iteration of the system. Table 2 contains

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the most important feedback received from the first usability study.

The second usability study was carried out with the second iteration of the games that were improved based on the first feedbacks. Both subjective and objective data were collected, either automatically from the system and in the cloud or directly by the experimenter. The first interesting analysis was made on the behavior of the system and its optimization model to determine if they produce the expected content. The number of the presentation for each piece of reading material (here called "presentation") and each wrong answer of the users (here called "error") were recorded for each item of reading material.

The data of the presented words (the number of presentations) was analyzed relative to the frequency of the words in the language. As explained previously, frequency was chosen as the value parameter of the words in the optimization model. These frequencies were classified into 53 levels. Since the games were using the same optimization model with minor tweaks, the data of the three games were combined to obtain more statistically significant data.

Table 2. The collected feedback from the first study.

Feedback	Action
The playing times of the games were not uniform.	The SD of time decreased from 331 s (M = 388) in the first experiment to 62 s (M = 477) in the second experiment.
The scores of the games were not uniform.	The SD of scores decreased from 393 (M = 525) in the first experiment to 347 (M = 626) in the second experiment.
The Thanksgiving game (AWSR task) was difficult to play.	– Multiple immediate visual and auditory feedbacks were added to the interactions. – The speed of the game was decreased. – A replay function was added for listening to the pronunciation of the words again.
The original task in the Easter game was not effective.	This task was completely removed and replaced by a task based on the Reading Accelerated Program [25].
Some of the reading materials were similar to the prior sessions.	It was intentionally designed in this way as blocked practice at the first stage of automaticity acquisition, particularly for young children, leads to better results [45].
The game (AWMR task) was difficult for English as a second language children.	This problem was not reported when testing with native English speakers in the second experiment.
Speech recognition in the Christmas game (AWD task) was not precise.	The speech recognition system was improved by limiting the recognition possibilities to only active word in the game scene.

It was expected that the system would start by selecting the words or word lists having the highest frequency level. However, because the numbers of words in each frequency level were not equal, the number of presentations should be moderated with this disparity to reach meaningful data. For this objective, the number of presentations for each frequency level was divided by the number of words (or word lists) existing at that level. The results are displayed in Figure 2, which looks obvious since users often played the game only once or twice.

Additionally, it was expected that the model would more often select the words having the highest frequency levels and less often the words with the next neighboring frequency levels. Similarly, the model prioritized the content with lower difficulty levels, which was marked by word length. Figure 3 illustrates this tendency. However, the number of presentations for each word length level had to be moderated by the number of words in that level. This is important because, for example, the only single-letter words in the English Language are "a" and "I." Therefore, the number of presentations in each word length level was divided by the number of words (or word lists) at that level.

Word length is not the only attribute that can be used as an indicator of word difficulty. For the investigation of this

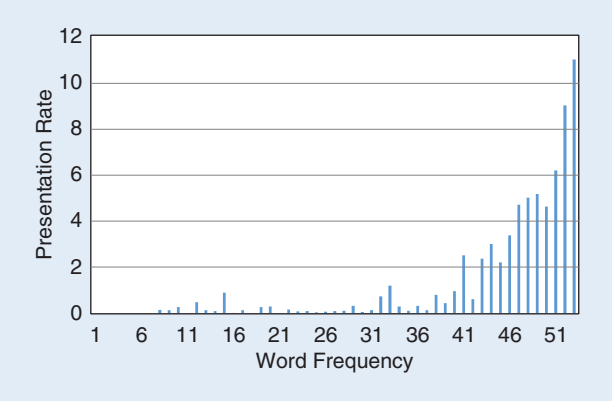


Figure 2. The presentation rate of the word frequency levels.

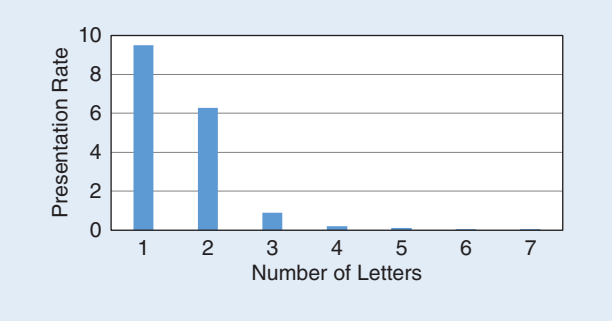


Figure 3. The presentation rate of the word length levels.

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choice, the data regarding the wrong answers (error rate) made by the users should be analyzed. To reach statistical significance, these automatically recorded data were combined across the three games that dealt with single words. As explained previously, the letter number was selected for the weight (difficulty) parameter in the optimization model. To evaluate this decision, the error rates (percentage of user errors in the number of presentations) for each letter number are presented in Figure 4. The trend line indicates that the errors increased with word length. However, as it appears in the diagram, the word length is not a perfect predictor of word difficulty. It is confirmed by the linear regression analysis that the letter number is not a statistically significant predictor of word difficulty

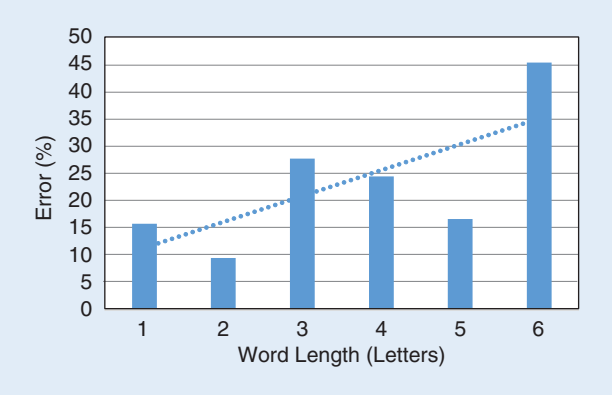


Figure 4. The percentage of errors for different word lengths.

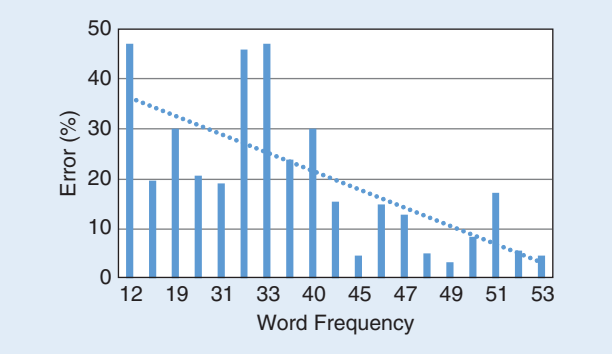


Figure 5. The percentage of errors for each frequency level.

Table 3. The situational motivation scale for the two usability studies.

Motivation Type	First Study	Second Study
Intrinsic motivation	77%	75%
Identified regulation	75%	70%
External regulation	37%	56%
Extrinsic regulation	40%	38%

($F(1,4) = 3.90$, $p = 0.119$), and it accounts for only 36.7% of the variability in error percentages. The lack of statistical significance in the linear regression model may be due to the low number of samples (in this case, the letter numbers).

Moreover, the same analysis was carried out for the frequency of words. The results demonstrated that the least frequent words produce more errors. Figure 5 portrays this inverse relationship between the word frequency and the percentage of errors. The results of the linear regression analysis indicated that word frequency is a statistically significant predictor of error percentage ($F(1,17) = 16.64$, $p = 0.001$). However, it only accounts for 46.5% of the variability in error percentages. Therefore, neither word length (R-squared = 36.7%) nor word frequency (R-squared = 46.5%) is a sufficiently accurate predictor of word difficulty. A combination of lexical information such as the word length, word frequency, and consistency of mapping between graphemes and phonemes as well as the age of the player is needed to build a more reliable word difficulty prediction model. It is also worth mentioning that for Figure 5, the frequency levels that had less than 20 presentations were discarded because they were too sparse to be reliable as an average of the error rates.

Furthermore, additional data were collected from the questionnaires presented to the users. The usability was assessed by the SUS test. Each question was represented by a score ranging from zero to four. Zero indicates the lowest usability, while four indicates the highest usability. The overall usability score was calculated for both experiments separately. The obtained scores were 75% for the first study and 80% for the second study. This increase was expected after improving the games according to the feedback received from the first study.

The second questionnaire in this study was the SIMS for the evaluation of the variables of intrinsic motivation, identified regulation, external regulation, and amotivation. The scores for each of these four variables were calculated and are presented in Table 3. In general, participants were more intrinsically motivated in the first study. The biggest gap between the first and the second study comes from the external regulation, which is significantly higher for the second usability study. This was expected as the average age of the participants in the second study (23.9) was significantly higher than the first study (14.4).

Finally, in the last questionnaire, the participants were asked to rate the games based on their level of difficulty and amusingness. The most difficult game was AWSR (Thanksgiving), with a rating of 3.75 out of 5, and the easiest game was APR (Easter), with a rating of 4.5. The least amusing game was AWD (Christmas), with a rating of 3.81, and the most amusing game was AWMR (Halloween), with a rating of 4.25.

Conclusion and Research Perspectives

Reading is an indispensable skill in modern life, and its importance cannot be overemphasized. Dyslexia as a

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specific learning difficulty is associated with problems of accuracy and speed in reading. The aspect of reading speed is often overlooked in the early grades and particularly in languages with opaque orthographies, such as English and French. In addition, vocabulary knowledge plays a crucial role in learning to read opaque languages, but technology-based vocabulary interventions have been truly neglected.

The objective of this research was to propose an intelligent game-based training program focused on the aspect of reading speed in the English language for young dyslexics. The design of the games was based on the self-efficacy and self-determination theories, and the training design was based on the principles of automaticity acquisition. Four minigames were designed that focus mainly on fostering automaticity of reading at the lexical level explicitly and the sublexical level implicitly as well as offering a basis for expanding vocabulary knowledge. An optimization model was developed and integrated into the games to allow for optimized content that maximizes the educational value while considering the individual abilities of each learner.

Two usability studies were conducted to validate the approach. The data collected from playing the games showed that the optimization model produced the expected content for the games. The content with the highest value and the lowest difficulty was preferred more frequently, and the next lower frequency and higher difficulty levels were selected less often. However, neither letter number nor word frequency was a perfect predictor of word reading difficulty. Hence, it is recommended to use a combination of lexical information for weighting the difficulty of words in the optimization model. The results of the SUS and SIMS are promising. The overall score of the usability test was 75% in the first study and 80% for the second design iteration of the games.

In future research, it is planned to use experimental data from the tests with students. These data will be used by the intelligent module of the games to determine the parameters of the optimization model with better accuracy. Additional games as well as customization options can be included in the games to enhance the feeling of autonomy for the users. Including emotion recognition in the interaction to tailor the games according to the user's emotional response or his/her facial expressions will also be considered. This will allow for taking into account the feelings of frustration or boredom during the gameplay.

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